WORKED SOLUTION

STEP I 1990, Question 13

Mechanics • Friction and Rigid Body Equilibrium

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Read the question before you go on. It is in the STEP Support Programme's question database:

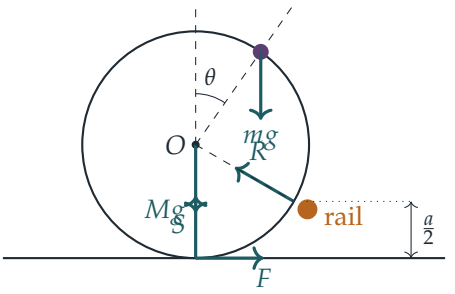
step.maths.org/questions/90-s1-q13

The key idea

Treat the particle first: it is an ordinary particle on a rough slope, the slope being the tangent plane to the cylinder at the point of contact. Then notice that, because the particle is in equilibrium, the total force it exerts on the cylinder is simply its weight, acting vertically down through the point of contact. Taking moments about the centre of the cylinder kills every force except that one and the friction from the plane, which is the quickest way in.

Solution

Model the particle as a point mass m and the cylinder as a uniform rigid cylinder of mass M whose centre of mass O is on its axis. The rail is smooth, so it exerts only a normal force; the plane and the cylinder's surface are rough with coefficient μ. Everything happens in the vertical cross-section through O, which we use from now on. Put the rail on the right.



The figure shows the forces on the cylinder (the mg at the particle is explained in the second part). Arrows are not to scale.

Equilibrium of the particle

At the point P of contact the outward normal makes angle θ with the upward vertical, so the tangent plane there is a slope inclined at θ to the horizontal. The forces on the particle are its weight mg, the normal reaction N along the outward normal and a friction force f along the tangent, up the slope. Resolving along and perpendicular to the normal,

$$N = mg\cos\theta, \qquad f = mg\sin\theta.$$

Friction can supply this provided $f \le \mu N$, that is,

$$\tan\theta \le \mu, \qquad \text{equivalently} \qquad \theta \le \arctan\mu.$$

(We need $0 \le \theta < 90^\circ$ for N to be positive, which holds as the particle is on top of the cylinder.)

Equilibrium of the cylinder once released

Ask yourself The particle presses on the cylinder with a normal force and a friction force. Do we need to handle them separately?

No. By Newton's third law the cylinder feels $-N$ and $-f$ at P. Since N and f together balance the particle's weight, their sum is mg vertically upwards, so the particle's total force on the cylinder is *mg vertically downwards, acting at P*. Its line of action is at horizontal distance $a\sin\theta$ from O, on the rail's side.

The other forces on the cylinder are:

- its weight Mg at O;
- the rail's reaction R, normal to the cylinder's surface and so directed along the radius towards O. The rail touches at height $a/2$, which is $a/2$ below O, so this radius makes 60° with the downward vertical. Hence R has a horizontal component $\frac{\sqrt{3}}{2}R$ away from the rail and an upward component $\frac{1}{2}R$;
- at the lowest point, a normal reaction S upwards and a friction force F horizontally (taken towards the rail).

Moments about O. The lines of action of Mg, R and S all pass through O. The particle's mg has moment $mg\,a\sin\theta$ (turning the cylinder towards the rail), and F acts at distance a below O, with the opposite sense. So

$$Fa = mg\,a\sin\theta \quad\Longrightarrow\quad F = mg\sin\theta.$$

This is positive, so friction does act towards the rail, as drawn. (Without the rail the cylinder would roll towards the rail's side; that is why the particle has to be on that side for the rail to be any use.)

Resolving horizontally: $\frac{\sqrt{3}}{2}R = F$, so

$$R = \frac{2mg\sin\theta}{\sqrt{3}} > 0,$$

consistent with the rail being able only to push.

Resolving vertically:

$$S = (M+m)g - \tfrac{1}{2}R = (M+m)g - \frac{mg\sin\theta}{\sqrt{3}},$$

which is positive since $\sin\theta < 1 < \sqrt{3}$ and $M > 0$, so the cylinder stays on the plane.

The plane can supply the friction provided $F \le \mu S$:

$$m\sin\theta \le \mu\left[(M+m) - \frac{m\sin\theta}{\sqrt{3}}\right] \iff m\sin\theta\left(\sqrt{3}+\mu\right) \le \sqrt{3}\,\mu\,(M+m).$$

So, once released, both bodies can remain in equilibrium provided $\tan\theta \le \mu$ and

$$\boxed{\sin\theta \le \frac{\sqrt{3}\,\mu\,(M+m)}{m\,(\sqrt{3}+\mu)}}$$

both hold. (No friction is needed at the smooth rail, and the friction between particle and cylinder was dealt with in the first part.)

The final part

With $m = 7M$ and $\mu = \frac{3}{4}$, the particle's condition is $\tan\theta \le \frac{3}{4}$, that is, $\sin\theta \le \frac{3}{5}$. The cylinder's condition is

$$\sin\theta \le \frac{\sqrt{3}\cdot\frac{3}{4}\cdot 8M}{7M\,(\sqrt{3}+\frac{3}{4})} = \frac{24\sqrt{3}}{7(4\sqrt{3}+3)} = \frac{24\sqrt{3}\,(4\sqrt{3}-3)}{7\cdot 39} = \frac{24(4-\sqrt{3})}{91}.$$

Which is the stricter? We have

$$\frac{24(4-\sqrt{3})}{91} < \frac{3}{5} \iff 120(4-\sqrt{3}) < 273 \iff 207 < 120\sqrt{3} \iff 42849 < 43200,$$

which is true. So the cylinder slipping on the plane is the binding constraint (only just), and the largest value of θ for equilibrium is

$$\theta_{\max} = \arcsin\frac{24(4-\sqrt{3})}{91} \approx 36.7^\circ,$$

just below $\arctan\frac{3}{4} \approx 36.9^\circ$.

Checks

- Limiting case $M \to \infty$: the cylinder's condition becomes vacuous, as a very heavy cylinder can always be held by friction; only $\tan\theta \le \mu$ remains.
- As $\theta \to 0$ the particle sits on top, its weight passes through O, and $F = R = 0$: no rail or friction is needed, as expected.
- Numerically, solving the three equilibrium equations from raw force vectors for 20 000 random choices of M, m, a, μ, θ agrees with the boxed condition every time; for $m = 7M$, $\mu = 0.75$ a fine scan of θ gives $\theta_{\max} \approx 36.737^\circ$, with $\sin\theta_{\max} \approx 0.59814$.

Common slips

- Taking the rail's force as horizontal. It is normal to the cylinder's surface, so it points along a radius, at 30° above the horizontal here.
- Forgetting the upward component $\frac{1}{2}R$ of the rail's force, which reduces S and so reduces the friction available from the plane.
- Stopping at $\theta = \arctan\frac{3}{4}$ without comparing with the cylinder's condition, which is (just) stricter.

Beyond STEP

The useful trick here, replacing the particle's contact forces by a single resultant equal to its weight, is a first glimpse of the idea that any system of forces on a rigid body reduces to a single force through a chosen point plus a couple. Statics of rigid bodies with several frictional contacts is a standard first-year topic, and the question of which contact slips first leads on to more subtle problems where the friction forces are not determined by the equations alone. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats forces, torques and rigid-body equilibrium carefully.

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